

Title: Employee Attrition Prediction

Vema Reddi Dondeti

Business Problem

Employee turnover is a significant challenge for organizations, leading to increased recruitment costs, training overhead, and disruption in operations. Losing high-performing employees can also impact morale and productivity. While exit interviews provide insights after the fact, predictive analytics offers a proactive solution.

This project aims to develop a predictive model that identifies employees likely to leave the organization. Early identification can enable HR departments to take preemptive actions, thereby improving retention rates and reducing associated costs. Using historical employee data, the model will classify employees into two categories: those likely to stay and those likely to leave.

Background/History

Employee attrition has been a long-standing issue in workforce management. Traditionally, companies relied on reactive strategies such as exit interviews and annual surveys to understand why employees leave. However, with the growing availability of HR data and advancements in machine learning, there is now a shift toward predictive analytics.

Attrition can be voluntary (resignations) or involuntary (terminations), and this project focuses on voluntary attrition. By analyzing patterns and characteristics of past attrition cases, organizations can identify at-risk employees and intervene appropriately.

Data Explanation

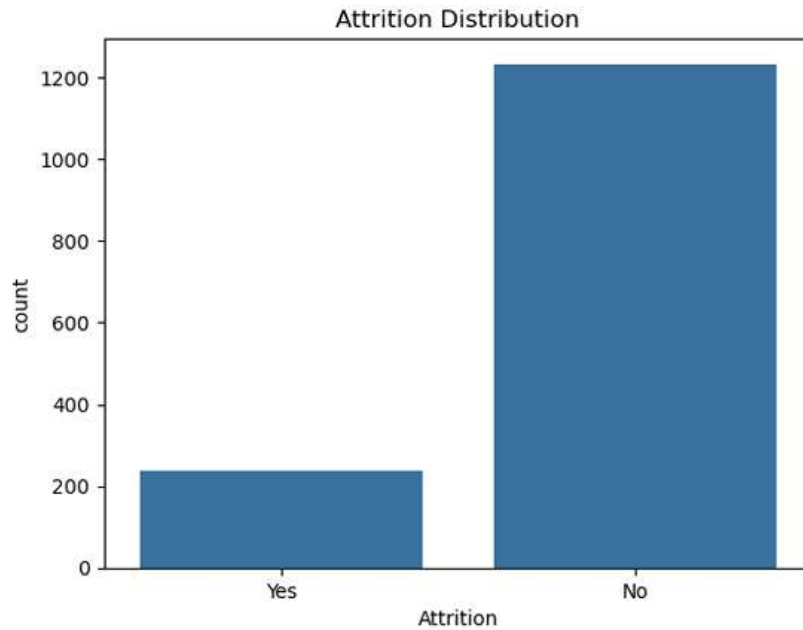
Dataset Description:

This project utilized the [IBM HR Analytics Employee Attrition & Performance](#) dataset, a publicly available dataset hosted on Kaggle. The dataset contains detailed information on 1,470 employees, with each entry representing an individual's demographic, job role, performance, and satisfaction attributes. Key features include age, job role, marital status, years at company, overtime status, job satisfaction, and more. The target variable is “Attrition,” indicating whether the employee left the organization. This rich dataset is well-suited for supervised machine learning techniques aimed at predicting employee turnover and uncovering the factors that contribute most to attrition.

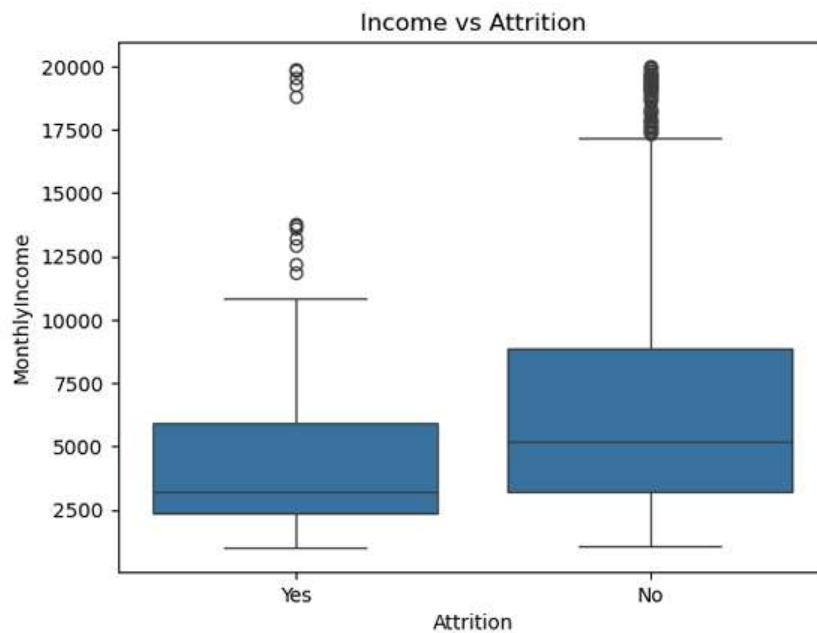
Exploratory Data Analysis (EDA)

Exploratory Data Analysis (EDA) was conducted to understand patterns within the employee dataset and identify key drivers of attrition. This step provided insights into the relationships between demographic, behavioral, and work-related features and employee turnover.

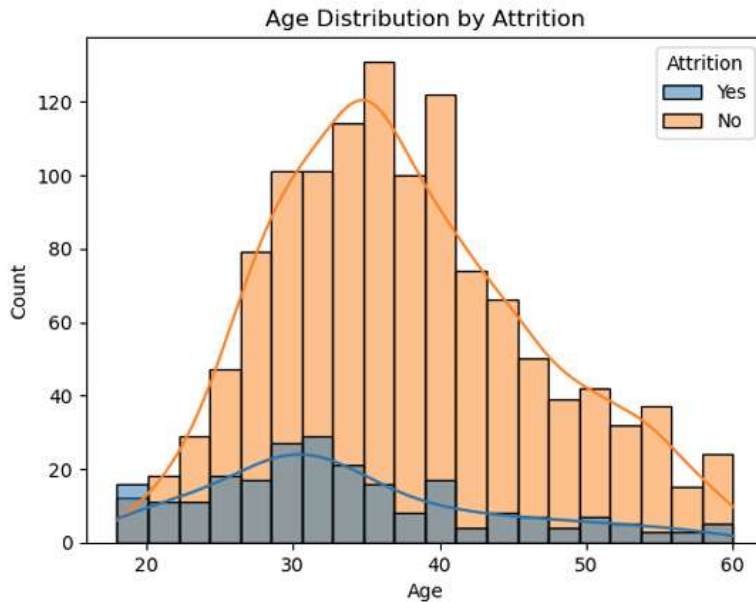
- **Attrition Distribution:** The target variable is imbalanced, with approximately 16% of employees labeled as having left the company. This skew was taken into account during the modeling phase through synthetic oversampling (SMOTE) to prevent bias toward the majority class.



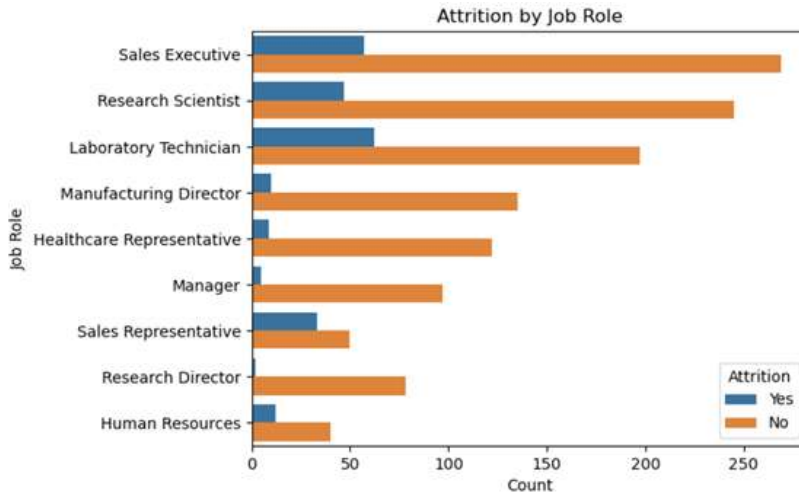
- **Income vs. Attrition:** Employees with lower monthly income showed a higher tendency to leave, suggesting compensation may play a role in attrition. However, the relationship is not strictly linear, indicating that income alone is not the sole driver.



- **Age Distribution by Attrition:** Employees who left the company tended to be younger, particularly in the 25–35 age range. Older employees exhibited higher retention, suggesting experience and tenure may contribute to long-term engagement.



- **Attrition by Job Role:** Certain roles—particularly Sales Representatives and Laboratory Technicians—showed significantly higher attrition rates compared to others like Managers and Research Directors. These differences may reflect variations in job demands, career progression, or satisfaction levels.



These visualizations highlighted potential predictive features and guided further feature engineering and model tuning.

Data Preparation

After exploratory analysis, irrelevant columns like EmployeeNumber, Over18, and StandardHours were dropped due to lack of variance or predictive value.

Missing values were confirmed to be absent. Categorical variables were encoded (e.g., one-hot encoding), and Attrition was mapped to binary values (Yes = 1, No = 0).

To address class imbalance, SMOTE was applied to oversample the minority class. Numerical features were standardized to ensure consistent scaling across the dataset.

Finally, the data was split into training and test sets using an 80:20 ratio for model evaluation.

Methods and Analysis

Three supervised models—Logistic Regression, Random Forest, and XGBoost—were trained on a SMOTE-balanced dataset to predict employee attrition. Logistic Regression was chosen for its interpretability, while Random Forest and XGBoost captured complex patterns.

Models were evaluated using accuracy, precision, recall, F1-score, and ROC-AUC. Logistic Regression showed the highest recall (77%) for attrition, making it effective for identifying employees likely to leave, though with lower precision (38%). Random Forest and XGBoost had higher overall accuracy (~85%) but low recall (<30%) for attrition, indicating bias toward the majority class.

Feature importance highlighted OverTime, YearsWithCurrManager, JobSatisfaction, and Age as key attrition predictors. ROC-AUC scores (~0.80) suggest good overall discrimination, with room for improving minority class detection.

Overall, Logistic Regression is recommended for proactive attrition prediction due to its better recall, helping HR teams target at-risk employees effectively.

Conclusion

This project successfully developed predictive models to identify employees at risk of attrition using supervised learning techniques. Among the models tested, Logistic Regression demonstrated the best ability to detect the minority class of employees likely to leave, offering valuable insights for HR intervention. Key factors such as overtime work, years with current manager, and job satisfaction were identified as significant predictors of attrition.

Overall, the project highlighted how predictive modeling can enable organizations to proactively address employee turnover, improve retention strategies, and maintain workforce stability. These

data-driven insights support targeted actions to enhance employee satisfaction and reduce costly attrition.

Assumptions

This analysis assumes that historical employee data accurately reflects future attrition trends and that all relevant features influencing attrition are included and correctly recorded. It also presumes that employee behavior and company policies remain stable over time. Additionally, the use of SMOTE for class balancing is assumed to improve model performance without introducing significant bias.

Limitations

The dataset is limited to one organization, which may reduce the generalizability of the findings. It does not account for external influences such as economic shifts or competing job offers. Some variables, like job satisfaction and engagement, are self-reported and may carry bias. Furthermore, the dataset lacks longitudinal data, limiting insight into how employee sentiment evolves over time.

Challenges

Key challenges included addressing class imbalance through synthetic sampling, which was necessary to improve detection of attrition cases. Interpreting complex models, particularly ensemble methods like XGBoost, posed difficulties in explaining predictions. Feature correlation and multicollinearity also had to be managed carefully. Lastly, tuning the balance between precision and recall was critical to minimize both false positives and false negatives.

Future Uses and Recommendations

Future uses of this model include incorporating external factors like economic trends and competitor activity to improve accuracy. Applying time-series methods could help predict not just who, but when employees might leave, enabling earlier interventions. Addressing ongoing class imbalance through advanced sampling or cost-sensitive learning can further enhance performance. Regular retraining with updated data will keep the model effective over time. Integrating this tool into HR systems will allow continuous monitoring of attrition risk, helping HR teams implement targeted retention strategies and improve workforce stability.

Implementation Plan

The attrition prediction model can be integrated into HR systems to routinely assess employee risk scores using current data. Dashboards can help HR prioritize interventions like workload management or mentoring for high-risk employees. Regular retraining with updated data will keep the model accurate. Clear policies should guide ethical use of these predictions to support employees positively.

Ethical Considerations

This project ensures employee data is anonymized and protected to maintain privacy. Predictions are used to assist, not penalize, employees and should be applied transparently. Efforts were made to reduce bias and handle class imbalance fairly. Compliance with data privacy laws like GDPR is essential to maintain trust and fairness in using predictive analytics.

References

IBM HR Analytics Employee Attrition & Performance. (2017, March 31). Kaggle.

<https://www.kaggle.com/datasets/pavansubhasht/ibm-hr-analytics-attrition-dataset>

scikit-learn: machine learning in Python — scikit-learn 1.7.0 documentation. (n.d.).

<https://scikit-learn.org/stable/>

Preet, A. (2024, November 11). Employee Attrition Prediction – A comprehensive guide.

Analytics Vidhya. <https://www.analyticsvidhya.com/blog/2021/11/employee-attrition-prediction-a-comprehensive-guide/>

Appendix A – Supplementary Materials

This appendix includes:

- **Code Summary:** Python scripts for data preprocessing, handling class imbalance with SMOTE, model training (Logistic Regression, Random Forest, XGBoost), evaluation metrics, and feature importance analysis.
- **Visualizations:**
 - Attrition class distribution
 - ROC Curves comparing model performance
 - Confusion Matrices for each model
 - Feature importance plots from all three models